

Price, Order, Activate: Mechanisms for Costly Verifiable Evidence in Collective Prediction

Abstract

Collective prediction often depends on costly evidence held by different agents. In cross-organizational cyber threat intelligence sharing, for example, one organization may observe DNS evidence, another endpoint telemetry, and a third registrar metadata; each artifact is checkable, but disclosure may require legal review, sanitization, analyst effort, or reputational risk. We study mechanism design for this setting when agents can reveal verifiable evidence or remain silent, but cannot fabricate. A principal aggregates disclosed evidence into a Bayesian forecast evaluated by a strictly proper scoring rule. The difficulty is that an artifact’s value is not fixed: it depends on what other organizations have already made public. We ask how a consortium should reward such disclosures. Our answer: **price additive evidence, order substitutes, and activate complements**. We first show that diffuse-credit rewards create a commons problem: evidence that is socially valuable can be privately unprofitable to disclose. We then introduce Sequential Scoring Rules (SSR), which pay each agent the score improvement generated by her disclosure, and prove local incentive compatibility. SSR implements first-best disclosure for every speaking order in a constant-additive benchmark, where each feature’s expected contribution is independent of both the public record and the feature realization. Outside this benchmark, marginal pricing fails in two canonical ways. Substitutes create order sensitivity: an expensive source can speak first and crowd out a cheaper equivalent source; cost-prioritized ordering repairs perfect-substitute classes. Complements create initiation deadlock: no agent’s first disclosure has positive marginal value although the bundle is valuable; Activation bonuses subsidize the first link, after which SSR completes the chain. We instantiate the framework in cyber threat intelligence examples and rational-agent stress tests across the additive, substitute, and complement regimes.

1 Introduction

Cyber threat intelligence (CTI) sharing provides a concrete instance of costly verifiable evidence disclosure. Organizations often observe different pieces of a security incident: DNS-resolution patterns, endpoint telemetry, certificate metadata, registrar information, malware hashes, or network-flow anomalies. Each artifact can be checked, timestamped, and shared with provenance. Each is also costly to disclose because it may require legal review, sanitization, customer-notification analysis, analyst time, or acceptance of reputational risk. Individually, these artifacts may be inconclusive. Together, they can move a consortium’s posterior on whether a domain is a command-and-control node, a binary is malicious, or a coordinated campaign is active.

Institutions such as CISA, ISACs, and ISAOs exist to aggregate such evidence, but under-sharing remains severe. The DHS Office of Inspector General reported in 2023 that active participants in CISA’s Automated Indicator Sharing program fell from 304 in 2020 to 135 in 2022, with submitted indicator volumes declining roughly tenfold [2]. The obstacle is not that disclosure is impossible. CISA’15 reduces liability and antitrust barriers, while STIX/TAXII and MISP reduce the cost of packaging and verifying evidence. The obstacle is that disclosure is *finite but costly*, while the informational benefit of one organization’s evidence is often shared by the whole consortium.

This gap creates a problem that is not simply one of forecasting. A consortium may know that its members hold potentially useful evidence, yet it cannot know in advance which artifacts they possess, what those artifacts will show, or how they will interact with the record already assembled. Each agent privately observes a verifiable feature X_i . An agent may reveal the true value of X_i or remain silent, but cannot fabricate. The principal updates a Bayesian posterior from the public record and is scored by a strictly proper scoring rule. Disclosure costs are borne by the agents. The question is: *how should the principal pay for evidence when the value of any one disclosure depends on what has already been revealed?*

This is not an ordinary prediction-market problem. In a prediction market, a participant changes a reported belief. Here, an organization decides whether to incur a cost to place a checkable artifact in the shared record. Proper scoring can reward an informational refinement once it has been made, but it does not by itself determine which costly artifacts should enter the record, or who should move first. What we show is that the correct institutional repair depends on the information structure: marginal pricing is enough for additive evidence, order control is needed for substitutes, and initiation subsidies are needed for complements.

1.1 A design rule

Our analysis leads to a very simple rule in practice: **Price additive evidence. Order substitutes. Activate complements.**

Information structure	SSR alone	Failure mode	Repair
Constant additive marginals	First-best	None	SSR
Perfect substitutes	Order-sensitive	Costly crowding out	Cost-prioritized ordering
Complements / k -way barriers	No revelation	Zero marginal value	Activation / activation sets
Type-dependent marginals	Forecast must use silence	Silence is informative	Strategic-inference model

Table 1: Taxonomy of marginal pricing: when SSR works, why it fails, and what repair is needed.

At the outset, it is useful to separate this setting from belief reporting. Agents cannot choose a posterior to announce; they can only decide whether to reveal a checkable artifact. That change creates the free-rider problem at once: under diffuse credit, an agent may withhold evidence whose social value exceeds its disclosure cost because she internalizes only a fraction of the forecast improvement (Theorem 1).

Once disclosure is understood as a costly refinement of the public record, SSR is the natural baseline: it pays the score improvement created by that refinement, and is locally incentive-compatible at every speaking turn (Theorem 2). In the constant-additive benchmark, where each feature’s marginal value is fixed across types and public records, SSR implements first-best disclosure for every speaking order (Theorem 3). This benchmark also clarifies the boundary of the model: once disclosure is type-dependent, silence itself becomes informative and the principal must reason about equilibrium non-disclosure (Proposition 1).

Once additivity fails, the difficulty separates into two different problems. Redundancy makes the order of speakers matter: a costly source can speak first and crowd out a cheaper equivalent (Theorem 4). When substitute classes are identifiable, cost-prioritized ordering restores first-best (Theorem 5). Complementarity leaves no one willing to move first: SSR can deadlock at no revelation even when positive welfare is possible (Theorem 6). Activation subsidizes the first link of a known complementary chain, after which SSR incentives complete the chain (Theorem 7); activation sets extend this to k -way barriers (Corollary 1). Sections 7–8 ground the theory in worked examples and computational simulations.

2 Related Work

The problem draws on several familiar literatures, but does not fit neatly within any one of them. It combines strategic experimentation, costly hard-evidence disclosure, scoring-rule incentives, and the institutional economics of CTI sharing. **Strategic experimentation.** Bolton and Harris [1] identify a free-rider effect in multi-player experimentation as a public good. Our free-riding theorem is the verifiable-disclosure analogue: information created by one agent benefits the group, but the disclosure cost is borne privately. Discreteness and verifiability remove the encouragement effect.

Contracting with externalities. Segal [18, 19] studies bilateral contracting with externalities. Activation has the same divide-and-conquer flavor: the principal pays an initiator bearing coordination cost, while downstream agents earn their marginal SSR contribution.

Evidence and disclosure games. Grossman [6] and Milgrom [15] establish full voluntary disclosure when evidence is free; Verrecchia [22] shows that positive costs break unraveling; Hart, Kremer, and Perry [9] characterize commitment outcomes. We study multiple evidence holders whose disclosures jointly determine a forecast, using a proper scoring rule to price each refinement.

Prediction markets and proper scoring. Hanson [7, 8] introduced market scoring rules (MSR), and Gneiting and Raftery [4] characterize strictly proper scoring rules. SSR uses Hanson’s payment identity but restricts the action space: an agent cannot choose a posterior; only reveal a verifiable feature or remain silent, making the problem costly subset selection rather than belief elicitation.

Submodular information acquisition. Krause and Guestrin [13] study sequential observation selection under submodularity; Singer [20] treats strategic suppliers under a budget. Our substitutes result is analogous: the speaking order can select the wrong costly representative.

CTI sharing economics. Gordon et al. [5], Hausken [10], Tosh et al. [21], Naghizadeh and Liu [16], Laube and Böhme [14], and Wagner et al. [23] document free-riding and adverse-selection problems in CTI sharing; the DHS OIG report [2] provides quantitative confirmation. NIST SP 800-150 [11] enumerates the disclosure cost taxonomy we use to interpret c_i .

3 Model

3.1 Environment

A finite set of agents $N = \{1, \dots, n\}$ jointly predict an unknown outcome $Y \in \mathcal{Y}$, finite. Nature draws $(X_1, \dots, X_n, Y)$ from a common-prior joint distribution μ with full support. Agent i privately observes x_i , the realization of X_i . Each agent owns a single feature; the multi-feature extension is straightforward. Feature i has a publicly known disclosure cost $c_i \geq 0$ borne by agent i upon disclosure, reflecting legal, compliance, competitive, and reputational components.

3.2 Disclosure operator and verifiability

Agent i ’s strategy is a disclosure function $\varphi_i : \mathcal{X}_i \rightarrow \{\text{disclose, withhold}\}$. *Verifiability:* any disclosed value equals the true realization—agents may withhold but not lie.

Remark (Scope of verifiability in CTI). We distinguish verifiable raw evidence (a signed log fragment, domain sighting with hash and timestamp, STIX object with provenance) from unverifiable interpretation (actor attribution, campaign-intent assessment). Our mechanism applies directly to the former. When partial verifiability holds, peer-prediction techniques [17, 24] can complement the scoring-rule layer.

3.3 Public record, forecast, and scoring

Let $A \subseteq N$ denote the disclosed set and x_A the disclosed values. We define the principal's forecast:

$$p_A(\cdot \mid x_A) := \Pr_\mu(Y = \cdot \mid X_A = x_A).$$

Throughout the main model, the public forecast is simply the posterior given the evidence that has actually been disclosed; it does not treat silence as a signal. This is without loss in our main positive benchmark, where disclosure decisions are type-independent, and on the equilibrium paths of our negative examples, where agents either always disclose or always withhold. When disclosure strategies are genuinely type-dependent, non-disclosure becomes informative; incorporating that signal requires a richer equilibrium model (§9).

A *strictly proper scoring rule* $S : \Delta(\mathcal{Y}) \times \mathcal{Y} \rightarrow \mathbb{R}$ assigns reward $S(p, y)$ to forecast p when outcome y is realized. Strict propriety means $\mathbb{E}_{Y \sim Q}[S(Q, Y)] > \mathbb{E}_{Y \sim Q}[S(P, Y)]$ for all $P \neq Q$. By Gneiting and Raftery [4], every strictly proper S admits a representation $S(P, y) = G(P) + \langle G'(P), \delta_y - P \rangle$ for strictly convex G , with $\mathbb{E}[S(P, Y) \mid Y \sim P] = G(P)$. Standard examples: the quadratic (Brier) rule $S(P, y) = 2p_y - \|P\|^2$, for which $G(P) = \|P\|^2$; and the logarithmic rule $S(P, y) = \log p_y$.

3.4 Two value functions: ex ante versus interim

The next distinction is easy to blur, but essential. $V(A)$ asks what a set of disclosures is worth before evidence is realized; $\bar{\delta}$ asks what an agent expects to gain after observing her own evidence. These are generally not the same quantity.

Definition 1 (Ex ante value). $V(A) := \mathbb{E}[S(p_A(\cdot \mid X_A), Y)]$ is the unconditional expected score of the Bayes-optimal forecast given disclosure of A . By construction, $V(\emptyset) = G(\mu_Y)$.

Definition 2 (Interim expected gain). Conditional on the current public record (I, x_I) and a candidate disclosure of feature set A with realized values x_A ,

$$\bar{\delta}(A \mid I, x_I, x_A) := \mathbb{E}_Y[S(p_{I \cup A}(\cdot \mid x_{I \cup A}), Y) - S(p_I(\cdot \mid x_I), Y) \mid x_{I \cup A}].$$

This is the expected score increment conditional on fully realized values. By strict propriety, $\bar{\delta} \geq 0$.

Lemma 1 (Averaging identity). For any $I \subseteq N$ and $A \subseteq N \setminus I$,

$$V(I \cup A) - V(I) = \mathbb{E}_{X_I, X_A}[\bar{\delta}(A \mid I, X_I, X_A)].$$

Pointwise equality $V(I \cup A) - V(I) = \bar{\delta}(A \mid I, x_I, x_A)$ does not hold in general; it holds for almost every realization only if the conditional gain is constant in (x_I, x_A) .

Proof. Tower property: $\mathbb{E}_{X_I}[\mathbb{E}_{X_A \mid X_I}[\bar{\delta}]] = \mathbb{E}[S(p_{I \cup A}, Y)] - \mathbb{E}[S(p_I, Y)] = V(I \cup A) - V(I)$. Pointwise equality fails generically because Bayes updates depend on the realization x_I . $\square$

3.5 Transfers, utilities, and welfare

When agent i_t discloses at step t , updating the public record from $\mathcal{I}_{t-1}$ to $\mathcal{I}_t$, she receives (ex post, after Y is realized) the SSR *transfer*

$$\tau_t := S(p_{\mathcal{I}_t}, Y) - S(p_{\mathcal{I}_{t-1}}, Y).$$

Her net *utility* is $u_{i_t} := \tau_t - c_{i_t}$. If she withholds, $\mathcal{I}_t = \mathcal{I}_{t-1}$ and both τ_t and u_{i_t} are zero. Transfers telescope: $\sum_t \tau_t = S(p_{\mathcal{I}_T}, Y) - S(p_{\mathcal{I}_0}, Y)$; utilities do not.

Budget feasibility. Under bounded scoring rules (Brier: $S \in [-1, 1]$) or clipped log score with range $[S_{\min}, S_{\max}]$, total transfer is bounded by $S_{\max} - S_{\min}$.

Define *incremental value* $\tilde{V}(A) := V(A) - V(\emptyset)$ and *incremental welfare*

$$W(A) := \tilde{V}(A) - \sum_{i \in A} c_i = V(A) - V(\emptyset) - \sum_{i \in A} c_i.$$

By construction, $W(\emptyset) = 0$. The *first-best* welfare is $W^* := \max_{A \subseteq N} W(A)$. We exclude transfers from welfare, as they are pure redistributions between principal and agents.

3.6 CTI mapping

Model object	CTI interpretation
Y ; settlement	Outcome (coordinated campaign? C2 node?); adjudicated post-incident
X_i ; verifiability	Local IOC, artifact, or log fragment; hash, timestamp, provenance metadata
c_i	Legal review, sanitization, confidentiality, competitive exposure, analyst time
Principal; budget	ISAC / CISA / consortium; dues, insurer, or public-sector sponsor
Activation bonus	Bounty for first credible artifact in a suspected complementary chain

Table 2: Model-to-CTI mapping.

Protocol sketch. In a stylized consortium implementation, the group fixes a settlement outcome and horizon (e.g., “domain confirmed malicious within 30 days”), members register verifiable evidence classes, and the principal announces the scoring rule, ordering policy, and any activation bonuses. Members then disclose signed STIX/MISP objects or withhold; after adjudication of Y , SSR payments and activation bonuses are settled. Settlement, verification, ordering, and budget are institutional choices; marginal pricing and activation are supplied by the mechanism.

Remark (Bounded-liability implementation). A realized score increment τ_t can be negative. Under any bounded scoring rule, however, the worst-case loss is finite, so the consortium can implement the same marginal incentives via a fixed participation payment or escrow large enough to offset it. Such fixed shifts preserve the SSR disclosure threshold (Theorem 2) because they do not depend on the disclosure action at the margin; by contrast, truncating negative increments feature-by-feature would generally distort incentives. We assume risk neutrality throughout.

4 Diffuse Credit and Sequential Scoring Rules

4.1 Free-riding under diffuse credit

The diffuse-credit mechanism splits the realized score gain equally: each agent receives $(1/n) \cdot [S(p_A, Y) - S(p_\emptyset, Y)]$ minus her own disclosure cost.

Theorem 1 (Free-riding under diffuse credit). *For every $n \geq 2$ there exists an instance with a feature i such that (a) disclosing i is welfare-improving— $\tilde{V}(\{i\}) > c_i$ —yet (b) the unique pure-strategy Nash equilibrium of the diffuse-credit mechanism withholds i .*

Proof. Agent 1 owns a single feature with positive informational value $v := \tilde{V}(\{1\}) > 0$. Agents 2, ..., n own degenerate features contributing 0 to $\tilde{V}$ in every subset. Choose $c_1 \in (v/n, v)$; set c_i prohibitively large for $i \geq 2$. Then $v - c_1 > 0$ (welfare-improving), but agent 1's payoff from disclosing under diffuse credit is $v/n - c_1 < 0$. Withholding strictly dominates. Since no other agent has an informative feature, the unique pure-strategy NE withholds feature 1. Equilibrium welfare is $0 < v - c_1 = W^*$. $\square$

4.2 Sequential Scoring Rules

A speaking order $\sigma = (i_1, \dots, i_n)$ is announced; at each turn t the active agent i_t observes the public record $(\mathcal{I}_{t-1}, x_{\mathcal{I}_{t-1}})$ and chooses to disclose or withhold. After all turns, Y is realized; agent i_t receives transfer τ_t and bears cost c_{i_t} if she disclosed.

Theorem 2 (SSR local incentive compatibility). *At her turn, agent i_t with type x_{i_t} and public record $(\mathcal{I}_{t-1}, x_{\mathcal{I}_{t-1}})$ maximizes her expected turn- t net utility by disclosing if and only if*

$$\bar{\delta}(\{i_t\} \mid \mathcal{I}_{t-1}, x_{\mathcal{I}_{t-1}}, x_{i_t}) > c_{i_t},$$

and is indifferent at equality.

Proof. By definition of $\bar{\delta}$, the agent's conditional expected transfer from disclosing is $\mathbb{E}_Y[S(p_{\mathcal{I}_t}, Y) - S(p_{\mathcal{I}_{t-1}}, Y) \mid x_{\mathcal{I}_t}] = \bar{\delta}(\{i_t\} \mid \mathcal{I}_{t-1}, x_{\mathcal{I}_{t-1}}, x_{i_t})$. Non-negativity $\bar{\delta} \geq 0$ follows because the refined posterior $p_{\mathcal{I}_t}$ is a mean-preserving refinement of $p_{\mathcal{I}_{t-1}}$ and G is strictly convex (conditional Jensen). Her expected net utility from disclosing is $\bar{\delta} - c_{i_t}$; from withholding, 0. The threshold rule follows. $\square$

Why this is not just a market scoring rule. In Hanson's MSR, an agent selects a posterior report and strict propriety ensures truthful elicitation. In SSR, the agent's only action is whether to pay a cost to refine the public record by revealing a verifiable feature. Proper scoring provides a local price for each refinement, but the global problem is decentralized selection of costly information subsets. The rest of the paper characterizes where this logic works, where it fails, and what repairs are needed.

Remark (Strategic delay). Theorem 2 establishes a single-turn best response. In the one-shot game where each agent is called exactly once—the natural CTI setting where each organization's analyst has finite time at the table—there is no delay. Multi-round extensions with stochastic termination are a natural direction (see §9).

4.3 First-best under constant additive marginals

Assumption 1 (Constant additive marginals). *For every agent i , every $I \subseteq N \setminus \{i\}$, every realization x_I , and every type x_i ,*

$$\bar{\delta}(\{i\} \mid I, x_I, x_i) = g_i$$

for a constant $g_i \geq 0$ that depends neither on the public record nor on the agent's realized type.

Assumption 1 is deliberately severe. It says that a feature has the same expected value no matter what has already been disclosed or what value that feature happens to take. This is strictly stronger than conditional independence of features given Y . Conditional independence yields factorized likelihoods, but the value of an additional signal in proper-score terms still typically depends on the current posterior's curvature. Assumption 1 is the cleanest benchmark under which SSR achieves first-best without additional control over order, activation, or inference from silence.

Why this is a baseline, not a description of ordinary signal structures. In a typical Bayesian problem, the value of a signal depends on the current posterior and on the signal itself. Assumption 1 deliberately removes both sources of variation, and the payoff is a clean baseline from which the paper’s negative results and repairs become meaningful.

Crucially, because g_i is constant in x_i , the disclosure decision $\varphi_i(x_i)$ is type-independent: agent i either always discloses (if $g_i \geq c_i$) or always withholds (if $g_i < c_i$). Withholding therefore carries no information about x_i beyond the prior, and the mechanical forecast of §3 coincides with the fully Bayesian posterior—no strategic inference from non-disclosure is needed.

Theorem 3 (First-best under constant additive marginals). *Under Assumption 1, for any speaking order σ and a fixed tie-breaking rule that selects disclosure at equality, SSR has the unique tie-broken SPE outcome $A^* = \{i : g_i \geq c_i\}$, achieving first-best welfare $W^* = \sum_{i: g_i \geq c_i} (g_i - c_i)$. Without tie-breaking, equilibrium outcomes can differ only on zero-surplus features with $g_i = c_i$, and all such outcomes achieve first-best welfare.*

Proof. Under Assumption 1, agent i ’s disclosure decision reduces by Theorem 2 to the threshold $g_i \geq c_i$, which depends neither on the public record nor on others’ strategies. Since each agent’s expected marginal contribution is a constant, the disclosed set is $\{i : g_i > c_i\}$ (plus tie-breaking) regardless of speaking order. Incremental welfare decomposes additively as $W(A) = \sum_{i \in A} (g_i - c_i)$, so the disclosed set maximizes welfare. Each agent’s strategy conditions only on her own constant g_i and cost c_i , so the SPE is unique up to ties. $\square$

4.4 The boundary of marginal pricing

Theorem 3 gives a sufficient condition. The two canonical departures from Assumption 1 exhibit the two failure modes of marginal pricing.

Proposition 1 (Canonical departures from additive marginal pricing). *Constant additive marginals (Assumption 1) are sufficient for order-invariant first-best implementation by SSR (Theorem 3). When constant additivity fails, each departure triggers a distinct failure mode:*

- (a) Non-additive $\tilde{V}$ with substitutes: *SSR becomes order-sensitive, with welfare ratios that vanish under adversarial speaking orders (Theorem 4).*
- (b) Non-additive $\tilde{V}$ with complements: *SSR deadlocks at no-revelation equilibria with zero welfare despite positive first-best (Theorem 6).*
- (c) Type-dependent additive marginals: *disclosure becomes type-contingent, so non-disclosure is informative about the agent’s type. The mechanical forecast of §3 then diverges from the fully Bayesian posterior that accounts for strategic withholding, introducing a distinct modeling complication outside the scope of the current framework.*

Failures (a) and (b) are proved in §5 via explicit constructions. Failure (c) identifies the boundary between the mechanical-forecast model and a richer strategic-inference model; a full treatment is left to future work (§9).

These cases do not exhaust every way SSR can fail. They isolate the two basic ways a locally correct price can still produce the wrong allocation: one artifact can crowd another out, or it can be worthless until another artifact is already public.

5 Limits of Marginal Scoring

The additive case is useful precisely because it is exceptional. Once evidence overlaps or works only in combination, a locally correct price need not select the right set of disclosures.

5.1 Monotonicity of V

Lemma 2 (Monotonicity). *For every $A \subseteq B \subseteq N$, $V(A) \leq V(B)$.*

Proof. Let $\sigma_A \subseteq \sigma_B$ be the σ -algebras generated by X_A and X_B . Bayes-rationality gives $p_A(y \mid \cdot) = \mathbb{E}[p_B(y \mid \cdot) \mid \sigma_A]$. By Gneiting–Raftery, $V(A) = \mathbb{E}[G(p_A)]$ for strictly convex G . Applying conditional Jensen: $G(p_A) = G(\mathbb{E}[p_B \mid \sigma_A]) \leq \mathbb{E}[G(p_B) \mid \sigma_A]$. Taking unconditional expectations: $V(A) = \mathbb{E}[G(p_A)] \leq \mathbb{E}[G(p_B)] = V(B)$. The proof uses convexity of G in the forecast and Bayes consistency $p_A = \mathbb{E}[p_B \mid \sigma_A]$. $\square$

5.2 Order sensitivity under substitutes

Theorem 4 (Order sensitivity). *There exists an instance with monotone submodular $\tilde{V}$ and modular costs c such that (a) first-best welfare $W^* = \Theta(1)$, and (b) under SSR with adversarial speaking order, the unique SPE outcome has welfare at most $\varepsilon \cdot W^*$ for any $\varepsilon > 0$ of one’s choosing.*

Proof. Two single-feature agents. Let $Y \sim \text{Bernoulli}(1/2)$. Feature 1 fully reveals Y : $X_1 = Y$. Feature 2 is a noisy reading with $\Pr[X_2 = Y \mid Y] = 1 - \alpha$ for some $0 < \alpha < 1/2$, conditionally independent of X_1 given Y . Since X_1 determines Y , feature 2 is conditionally redundant given feature 1: $\bar{\delta}(\{2\} \mid \{1\}, x_1, x_2) = 0$ pointwise. Under the normalized quadratic score (rescaled by 2, with $\tilde{V}(\emptyset) = 0$): $\tilde{V}(\{1\}) = 1$, $\tilde{V}(\{2\}) = 1 - \eta$, $\tilde{V}(\{1, 2\}) = 1$, where $\eta := 4\alpha(1 - \alpha) \in (0, 1)$. Agent 1’s interim marginal at the empty record equals 1 deterministically (each realization fully reveals Y).

Set costs $c_1 = 1 - \delta$, $c_2 = \delta$, with $0 < \delta < \min\{\eta/2, (1 - \eta)/2\}$. First-best discloses $\{2\}$: $W(\{2\}) = (1 - \eta) - \delta > 0$ (positive since $\delta < (1 - \eta)/2$). Under SSR with adversarial order (agent 1 first): $1 > c_1 = 1 - \delta$, so agent 1 discloses (Theorem 2); then $\bar{\delta}(\{2\} \mid \{1\}, x_1, x_2) = 0 < \delta = c_2$, so agent 2 withholds. SPE welfare: $W(\{1\}) = 1 - (1 - \delta) = \delta$. The ratio $\delta / [(1 - \eta) - \delta]$ vanishes as $\delta \rightarrow 0$. $\square$

Remark. SSR gets each local decision right and still gets the allocation wrong. The pathology is that an expensive substitute is disclosed first under marginal pricing, crowding out a cheaper source.

5.3 Cost-prioritized ordering repairs perfect substitutes

When the principal can identify substitute classes, ordering by cost restores efficiency.

Definition 3 (Perfect-substitute classes). *Features partition into classes $G_1, \dots, G_m$. The interim marginals satisfy: (a) within-class redundancy: $\bar{\delta}(\{i'\} \mid I, x_I, x_{i'}) = 0$ pointwise whenever $I \cap G_j \neq \emptyset$ for $i' \in G_j$; (b) constant first-disclosure marginal: $\bar{\delta}(\{i\} \mid I, x_I, x_i) = v_j > 0$ pointwise whenever $I \cap G_j = \emptyset$ for $i \in G_j$; (c) cross-class independence: the first-disclosure marginal v_j of class G_j does not depend on which features from other classes have been disclosed. Let $c_{\min, j} := \min_{i \in G_j} c_i$. Ties among equal-cost members within a class are broken by a fixed deterministic rule (e.g., lexicographic).*

Theorem 5 (Cost-prioritized SSR repairs perfect substitutes). *Suppose features have the perfect-substitute class structure of Definition 3 and the principal calls speakers in cost-ascending order within each class (with the deterministic tie-breaking rule of Definition 3). Under a fixed tie-breaking convention that selects disclosure for the cheapest first representative when $v_j = c_{\min, j}$, and withholding for residual representatives with zero marginal value, SSR has the unique tie-broken SPE outcome: a cheapest representative of class G_j discloses if and only if $v_j \geq c_{\min, j}$, while all remaining same-class representatives withhold. The resulting welfare is $\sum_{j: v_j > c_{\min, j}} (v_j - c_{\min, j})$.*

Proof. Within class G_j under cost-ascending order, the cheapest agent i^* is called first (with ties broken deterministically). By (b), her interim marginal at any record I with $I \cap G_j = \emptyset$ equals v_j . If $v_j > c_{\min,j}$, Theorem 2 makes disclosure strictly optimal; if $v_j < c_{\min,j}$, withholding is strict; and if $v_j = c_{\min,j}$, the stated tie-breaking convention selects disclosure. After the first disclosure, residual agents with $c_i > 0$ strictly withhold, while zero-cost residual agents withhold under the same convention. By (c), within-class outcomes are independent of cross-class disclosures, so the argument applies to each class independently. The realized disclosed set is the cheapest representative of each profitable class, which is first-best. $\square$

Remark. Class structure is often observable in CTI settings (features tagged by sector, artifact type, or STIX category), making cost-prioritized ordering operationally feasible.

Approximate substitutes. Exact substitutes are a useful starting point because they make the ordering logic transparent. The next proposition asks what remains when additional members of a class retain a small residual value.

Proposition 2 (Representative selection). *Suppose $\tilde{V}(A) = \sum_{j=1}^m f_j(A \cap G_j)$, with $f_j(\emptyset) = 0$, $f_j(\{i\}) = v_j$ for all $i \in G_j$, and $v_j \leq f_j(T) \leq v_j + (|T| - 1)\varepsilon_j$ for every nonempty $T \subseteq G_j$. Assume $\bar{\delta}(\{i\} \mid I, x_I, x_i) = v_j$ for all $i \in G_j$ whenever $I \cap G_j = \emptyset$. Consider representative selection: call agents in nondecreasing cost order within each class; continue past refusals; remove the class after the first disclosure. Then $W_{\text{rep}} \geq W^* - \sum_{j=1}^m (|G_j| - 1)\varepsilon_j$.*

Proof. Per class G_j , the mechanism obtains $W_j^{\text{rep}} = \max\{0, v_j - c_{\min,j}\}$. Any nonempty first-best set $T \subseteq G_j$ yields class welfare at most $v_j + (|G_j| - 1)\varepsilon_j - c_{\min,j}$, so $W_j^* \leq \max\{0, v_j - c_{\min,j} + (|G_j| - 1)\varepsilon_j\} \leq W_j^{\text{rep}} + (|G_j| - 1)\varepsilon_j$. Summing over classes gives the result. $\square$

This is an additive-loss guarantee for class-separable approximate substitutes, not a general approximation result for arbitrary submodular information.

5.4 Deadlock under complements

Theorem 6 (No-revelation deadlock). *There exists an instance with two single-feature agents such that $\tilde{V}(\{1\}) = \tilde{V}(\{2\}) = 0$, $\tilde{V}(\{1, 2\}) > 0$, $c_1 + c_2 < \tilde{V}(\{1, 2\})$, and the no-revelation profile is the unique SPE outcome of SSR. SPE welfare is 0; first-best welfare $W(\{1, 2\}) = \tilde{V}(\{1, 2\}) - c_1 - c_2 > 0$.*

Proof. Canonical noisy-parity construction. Let $X_1, X_2 \in \{0, 1\}$ be independent uniform; $Y = X_1 \oplus X_2 \oplus \varepsilon$ with $\varepsilon \sim \text{Bernoulli}(0.1)$ independent. Then Y is independent of either X_1 alone or X_2 alone, so $\tilde{V}(\{1\}) = \tilde{V}(\{2\}) = 0$. But $\tilde{V}(\{1, 2\}) > 0$ because the parity $X_1 \oplus X_2$ reveals Y up to ε .

At any speaking order under SSR: at turn 1, the active agent faces the empty record. By Lemma 1, $\mathbb{E}_{X_i}[\bar{\delta}(\{i\} \mid \emptyset, X_i)] = \tilde{V}(\{i\}) = 0$. Since $\bar{\delta} \geq 0$ (strict propriety) and has zero expectation, $\bar{\delta}(\{i\} \mid \emptyset, x_i) = 0$ for a.e. x_i . By Theorem 2, the active agent withholds (cost $c_i > 0$). With nothing disclosed, the turn-2 agent faces the same empty record and zero marginal: she also withholds. The unique SPE outcome is no revelation. $\square$

The k -way generalization is immediate: for k -way noisy parity, $\tilde{V}(S) = 0$ for all $|S| < k$, but $\tilde{V}(\{1, \dots, k\}) > 0$. SSR deadlocks for any k , any speaking order.

6 Activation as An Initiator Bonus

The complement case fails for a different reason than substitutes. The first artifact is not under-priced: by itself it is genuinely worth nothing. Once it is public, however, the rest of the chain can become privately worthwhile. The problem is that no one may be willing to move first.

6.1 Mechanism

Fix a chain $C = (1, 2, \dots, k)$ in which feature 1 is the designated initiator. The Activation mechanism augments SSR with a single transfer: pay agent 1 a bonus $b > 0$ if and only if she discloses X_1 at her turn in the chain order. All other turns use unmodified SSR transfers.

6.2 Chain completion

Theorem 7 (Chain completion via Activation). *Fix a chain $C = (1, \dots, k)$ and suppose:*

- (i) Profitable downstream links: *for every $r \in \{2, \dots, k\}$ and a.e. realization of $(x_1, \dots, x_{r-1})$,*

$$\bar{\delta}(\{r\} \mid \{1, \dots, r-1\}, x_{1:r-1}, x_r) > c_r \quad \text{for a.e. } x_r;$$

- (ii) Chain-respecting speaking order: *agents are called in order $(1, 2, \dots, k)$;*
- (iii) Strict initiation bonus: *$b > \max_{x_1} [c_1 - \bar{\delta}(\{1\} \mid \emptyset, x_1)]$ (the worst-case initiation gap plus a strict margin).*

Then under Activation + SSR, full chain disclosure is the unique SPE outcome. If additionally the chain is socially profitable— $W(C) = \tilde{V}(C) - \sum_{r=1}^k c_r > 0$ —then Activation implements a welfare-improving disclosure. The principal's expected expenditure is $b + \tilde{V}(C)$.

Proof. Backward induction along the chain. *Turn k :* agent k faces public record $\{1, \dots, k-1\}$ with values $x_{1:k-1}$; by (i), $\bar{\delta} > c_k$, so disclosure is strictly optimal (Theorem 2). *Turn $r < k$:* by the inductive hypothesis, all agents $r+1, \dots, k$ disclose at their turns. Agent r 's expected turn- r net utility from disclosure is $\bar{\delta}(\{r\} \mid \{1, \dots, r-1\}, x_{1:r-1}, x_r) - c_r > 0$ by (i); disclosure is strictly optimal. *Turn 1:* agent 1's expected payoff from disclosing is $\bar{\delta}(\{1\} \mid \emptyset, x_1) + b - c_1 > 0$ by (iii). Disclosure is strictly optimal for every realization x_1 . The unique SPE outcome is full chain disclosure. Welfare and expenditure follow by direct accounting. $\square$

6.3 Activation sets for k -way barriers

For k -way complementarities where every singleton has zero marginal (e.g., k -way parity), no single-initiator bonus suffices.

Corollary 1 (Activation sets). *Fix an ordered activation set $H = (h_1, \dots, h_m)$ and a residual chain $C \setminus H$. Suppose: (i) Activation bonuses cover gaps: for each h_j , set $b_{h_j} = \max_{x_{\mathcal{I}_{< j}}, x_{h_j}} [c_{h_j} - \bar{\delta}(\{h_j\} \mid \mathcal{I}_{< j}, x_{\mathcal{I}_{< j}}, x_{h_j})]^+ + \varepsilon_{h_j}$ for arbitrarily small $\varepsilon_{h_j} > 0$, where $[z]^+ := \max\{z, 0\}$; or, more cheaply, prefix-contingent bonuses $b_{h_j}(x_{\mathcal{I}_{< j}}) = \max_{x_{h_j}} [c_{h_j} - \bar{\delta}(\{h_j\} \mid \mathcal{I}_{< j}, x_{\mathcal{I}_{< j}}, x_{h_j})]^+ + \varepsilon_{h_j}$. (ii) Residual chain strictly profitable: once H is fully public, every residual link has $\bar{\delta} > c_r$ along the chain order. Then full disclosure is the unique SPE outcome. Under the fixed-bonus implementation, total realized activation expenditure is at most $\sum_{h \in H} c_h + \sum_h \varepsilon_h$, since each worst-case gap is at most c_h because $\bar{\delta} \geq 0$; the prefix-contingent implementation weakly lowers expenditure.*

Proof. Backward induction as in Theorem 7, first on the residual chain, then the activation set. $\square$

6.4 Discussion

Activation affects only the first move. A posted bonus brings the first artifact onto the table; standard SSR payments sustain the rest of the chain. Implementation needs no dynamic schedules, budget evolution, or stochastic termination. The principal should offer activation bonuses only for chains whose expected informational value exceeds total disclosure costs. Bonuses implement such chains but do not make an inefficient chain socially worthwhile. The cost of this minimalism is that the principal must know the chain structure, the initiator’s worst-case gap, and which downstream agents retain profitable interim marginals once the chain prefix is public. In CTI settings, this knowledge may be partly inferred from sector roles and known artifact-class observability (e.g., a registrar is usually closest to a domain-resolution chain).

Simply contract for the evidence? A consortium may know which organizations occupy different evidentiary roles (e.g., registrars, vendors, or service providers) without knowing who has relevant evidence in a particular incident, what the evidence will contain, or how far it will move the posterior. Ordering and activation use that coarse structural information to reduce free-riding and coordination failure without requiring the principal to know realized evidence *ex ante*.

7 CTI-Style Examples

Before turning to randomized instances, it is useful to see the mechanism in three small cases. For simplicity, we choose a deliberately simple quadratic score $S(p, y) = 2p_y - \|p\|^2$, so that the source of each incentive is clear.

7.1 Regime A: Constant additive benchmark

Suppose three organizations each hold an independent indicator whose expected contribution is 0.10, regardless of what else is public. Formally, $\bar{\delta}(\{i\} \mid I, x_I, x_i) = g_i = 0.10$ for all I, x_I, x_i , with costs $c_i = 0.05$. By Theorem 3, SSR discloses all three under any speaking order; $W = 0.15 = W^*$. Under diffuse credit ($n = 3$), each agent internalizes only $0.033 < 0.05$; the unique NE withholds all three (Theorem 1); welfare is 0.

7.2 Regime B: Substitutes (order sensitivity)

Suppose one organization holds a definitive but expensive reading, while another holds a cheaper noisy reading of the same event. Formally: $X_1 = Y$ (full revelation); X_2 noisy with error $\alpha = 0.2$, yielding $\eta = 0.64$. Under the normalized quadratic score: $\tilde{V}(\{1\}) = 1$, $\tilde{V}(\{2\}) = 0.36$, $\tilde{V}(\{1, 2\}) = 1$. Costs $c_1 = 0.98$, $c_2 = 0.02$. First-best discloses $\{2\}$; $W^* = 0.34$. Adversarial order (agent 1 first): agent 1 discloses ($1 > 0.98$), crowding out agent 2 ($0 < 0.02$); $W = 0.02$. Cost-ascending order (agent 2 first): agent 2 discloses ($0.36 > 0.02$), agent 1 is redundant; $W = 0.34 = W^*$.

7.3 Regime C: Complements with Activation

Suppose an endpoint event and a network trace are each uninformative alone but diagnostic together. We model this as a parity structure: $X_1, X_2 \in \{0, 1\}$ independent uniform; $Y = X_1 \oplus X_2 \oplus \varepsilon$, $\varepsilon \sim \text{Bernoulli}(0.1)$. Under the quadratic score: $\tilde{V}(\{1\}) = \tilde{V}(\{2\}) = 0$; $\tilde{V}(\{1, 2\}) = 0.32$; costs $c_1 = c_2 = 0.05$. SSR alone deadlocks (Theorem 6): $W = 0$; first-best $W^* = 0.22$. With Activation (chain $C = (1, 2)$, bonus $b = 0.06$), agent 1’s payoff from disclosing is $0.01 > 0$; once X_1 is public, agent 2’s marginal is $0.32 > 0.05$ for every realization. Full disclosure is implemented; $W = 0.22 = W^*$; principal expenditure 0.38.

8 Computational Experiments

The preceding examples are intentionally sharp. We next randomize within the same three structural families to ask whether the qualitative distinctions survive away from a single hand-picked set of costs and values. Each instance specifies a reduced-form incremental expected-score value function $\tilde{V}$, publicly known costs, and the relevant mechanism; first-best welfare is computed by exhaustive subset enumeration. For order-sensitive mechanisms, welfare ratios are averaged over 50 uniformly random speaking orders. These simulations test the mechanism logic under randomized inputs; they do not model how organizations or people actually behave.

Instance families. *Additive:* 5 features, $g_i \sim U(0.05, 0.30)$, $c_i \sim U(0.01, 0.20)$. *Substitutes:* 6 features in 3 perfect-substitute classes, values $U(0.10, 0.40)$, costs $U(0.01, 0.30)$. *Complements:* 5 features with a 3-way complement chain ($\tilde{V} = 0.40$, zero on proper subsets), two strictly unprofitable additive decoys, activation set of size 2.

Mechanism	Additive	Substitutes	Complements
Diffuse credit	0.28 (0.02)	0.28 (0.02)	0.00 (0.00)
SSR (random order)	1.00 (0.00)	0.82 (0.01)	0.00 (0.00)
SSR + substitute ordering	1.00 (0.00)	1.00 (0.00)	0.00 (0.00)
SSR + regime-specific repair	1.00 (0.00)	1.00 (0.00)	1.00 (0.00)

Table 3: Mean welfare ratio to first-best (SE in parentheses) across 300 random instances. Diffuse credit is solved by backward induction. Seed, script, and all generated outputs are in the supplement.

Randomization changes the welfare ratios, but not the diagnosis. Diffuse credit underperforms across regimes because agents internalize only a fraction of the score value. SSR is first-best in the additive family but loses $\sim 18\%$ on substitutes when random ordering lets a costly representative crowd out a cheaper one. Cost-prioritized ordering fully recovers in the substitute family. SSR deadlocks in the complement family; activation sets restore first-best.

9 Discussion and Conclusion

SSR answers the local question of what a disclosure is worth, given what is already public. In this paper, we have asked the question of when that local answer is enough. It is enough under additivity; substitutes require control over order, and complements require someone to move first.

Our analysis here makes three simplifying assumptions: (1) the forecast conditions on disclosed evidence only, not on silence (it is exact under constant marginals, but requires a full equilibrium-belief model under type-dependent disclosure); (2) costs are assumed to be common knowledge; (3) risk neutrality, assuming limited liability calls for bounded payment variants.

Implementation of our scheme requires a settlement outcome and a budget source. Bounded scoring rules can cap SSR payments at $S_{\max} - S_{\min}$; and escrow or fixed participation payments can preserve threshold incentives for limited-liability participants.

In conclusion, costly verifiable disclosure is not an ordinary evidence game or a free-form belief market. The central difficulty is getting organizations to bear the cost of making evidence checkable when the value of each artifact depends on what is already public. SSR supplies the local price; ordering and Activation supply what local prices alone cannot: *price additive evidence*, *order substitutes*, and *activate complements*. Future work can explore and extend these ideas to scenarios involving delay, shared ownership, private costs, and partially verifiable claims.

References

- [1] P. Bolton and C. Harris. Strategic experimentation. *Econometrica*, 67(2):349–374, 1999.
- [2] DHS Office of Inspector General. CISA made limited progress in improving information sharing under the Cybersecurity Act of 2015. OIG-23-39, 2023.
- [3] Y. Du, S. Li, A. Torralba, J. B. Tenenbaum, and I. Mordatch. Improving factuality and reasoning in language models through multiagent debate. In *ICML*, 2024.
- [4] T. Gneiting and A. E. Raftery. Strictly proper scoring rules, prediction, and estimation. *J. Amer. Statist. Assoc.*, 102(477):359–378, 2007.
- [5] L. A. Gordon, M. P. Loeb, and W. Lucyshyn. Sharing information on computer systems security: An economic analysis. *J. Accounting and Public Policy*, 22(6):461–485, 2003.
- [6] S. J. Grossman. The informational role of warranties and private disclosure about product quality. *J. Law and Economics*, 24(3):461–483, 1981.
- [7] R. Hanson. Combinatorial information market design. *Information Systems Frontiers*, 5(1):107–119, 2003.
- [8] R. Hanson. Logarithmic market scoring rules for modular combinatorial information aggregation. *J. Prediction Markets*, 1(1):3–15, 2007.
- [9] S. Hart, I. Kremer, and M. Perry. Evidence games: Truth and commitment. *Amer. Econ. Rev.*, 107(3):690–713, 2017.
- [10] K. Hausken. Information sharing among firms and cyber attacks. *J. Accounting and Public Policy*, 26(6):639–688, 2007.
- [11] C. Johnson, L. Badger, D. Waltermire, J. Snyder, and C. Skorupka. Guide to cyber threat information sharing. NIST SP 800-150, 2016.
- [12] E. Kamenica and M. Gentzkow. Bayesian persuasion. *Amer. Econ. Rev.*, 101(6):2590–2615, 2011.
- [13] A. Krause and C. Guestrin. Near-optimal observation selection using submodular functions. In *AAAI*, pages 1650–1654, 2007.
- [14] S. Laube and R. Böhme. Strategic aspects of cyber risk information sharing. *ACM Computing Surveys*, 50(5):77:1–77:36, 2017.
- [15] P. R. Milgrom. Good news and bad news: Representation theorems and applications. *Bell J. Economics*, 12(2):380–391, 1981.
- [16] P. Naghizadeh and M. Liu. Inter-temporal incentives in security information sharing agreements. In *AAAI Workshop on AI for Cyber Security*, 2016.
- [17] D. Prelec. A Bayesian truth serum for subjective data. *Science*, 306(5695):462–466, 2004.
- [18] I. Segal. Contracting with externalities. *Quart. J. Econ.*, 114(2):337–388, 1999.
- [19] I. Segal. Coordination and discrimination in contracting with externalities: divide and conquer? *J. Econ. Theory*, 113(2):147–181, 2003.
- [20] Y. Singer. Budget feasible mechanisms. In *FOCS*, pages 765–774, 2010.
- [21] D. K. Tosh, S. Sengupta, C. A. Kamhoua, K. A. Kwiat, and A. Martin. An evolutionary game-theoretic framework for cyber-threat information sharing. In *IEEE ICC*, pages 7341–7346, 2015.

- [22] R. E. Verrecchia. Discretionary disclosure. *J. Accounting and Economics*, 5:179–194, 1983.
- [23] T. D. Wagner, K. Mahbub, E. Palomar, and A. E. Abdallah. Cyber threat intelligence sharing: Survey and research directions. *Computers & Security*, 87:101589, 2019.
- [24] J. Witkowski and D. C. Parkes. A robust Bayesian truth serum for small populations. In *AAAI*, pages 1492–1498, 2012.